Christopher Johnson

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Professional Summary

Software Engineer with 6 years of professional experience leading, owning, and developing novel systems to solve critical business problems at a management service operations startup targeting the psychiatry vertical. Designed scalable backend infrastructure at Google and developed and optimized hardware algorithms and the software to analyze the performance of the hardware for the graphics hardware division at Intel. Strong skills in creating software solutions for critical business problems, backend software development, algorithm design, and mathematical solutions with ability to work across multiple problem domains.

Employment

2025/11 – 2026/8

Software Engineer. Blossom Health.

- Built and owned the psychiatrist onboarding pipeline converting what was once hour long, intermittent, human process into status flips on an operations dashboard by developing a multi-stage integrations pipeline that reflected existing hiring operations. Unlocking scalable psychiatry hiring processes for the Blossom Health Psychiatry platform.
- Built and owned a insurance payer enrollment mirror with at most 1 day staleness for psychiatrists on the platform. Completely eliminating the need for what was once a tedious multi-hour error prone human sweep every few weeks to update psychiatrist enrollment data on all RCM integration platforms.
- Collaborated with multiple teammates to build a state-of-the-art patient facing Langgraph customer support agent that scheduled appointments, uploaded insurance, answered billing questions, and created medical documents across SMS, email, and internal app chat platform. Parallelized offline evals for this agent to remove time constraints on the development process.
- Scoped, developed, and owned the system automating Prior Authorizations. Completely revolutionizing the existing procedure from initiating prior authorizations via offshore manual labor when the patient is angry and without their medication, to pro-actively creating prior authorizations upon psychiatrist written prescription by leveraging LLM technology, document intelligence, traditional software infrastructures and browser automation. First obtained a deep understanding of the current state of payer prior authorizations across all payers including submission via documents as well as electronic portal systems. Then did research on what the most relevant document intelligence (AWS textract) and browser automation technology (Playwright) for this purpose was. Invented new ways to perform AI assisted document filling by combining both AWS textract output and leveraged LLMs to generate logic graphs for intelligently choosing which parts of the form to fill and what to fill for both inbound and outbound prior authorizations documents. Created a web portal agent using Playwright to login, choose a case, and answer questions about the prescription using a psychiatrist locked chart note then attaching this agent to a sweeper cron job using a traditional queue (Bullmq) to do multiple sweeps of each case to answer follow up questions until the prior authorization reached a terminal approved or denied status.

Employment (continued)

2022 – 2025/7 **Software Engineer.** Google LLC.

- Introduced caching technology into the Google Play Store architecture for user location fetching in C++, resulting in a significant 15% reduction in latency for Play Store queries. Identified the temporal patterns for user location data, chose and modified the most suitable cache system from a suite of options, then integrated caching mechanism into user location fetching system followed by fine tuning of the cache time to live and cache size to achieve optimal performance.
- Redesigned and implemented Play Store family location fetching software system employing design patterns such as optimized caching, concurrency, and event-driven architecture, resulting in an overall query reduction of 10% and fetching latency decrease by 7%. Using C++ and Java, software development of different services deployed on multiple servers was carried out to build this system. A one-shot offline pipeline was created to backfill data to avoid breaking production at a massive scale.
- Redesigned and implemented device data fallback system for user's context data in C++ to enable Google Play Store compliance with DMA regulation issues by the European Union, preventing potential penalties imposed by regulatory agencies. This was a year long effort resolving complex inter-dependencies among numerous software services within and outside the Google Play Store stack, impacting domains from app installs to out-of-box experiences of consumer devices around the world.
- Restructured and upgraded User Context Data fetching system for gateway servers to the Google Play Store using Java. This was part of a wider effort to upgrade and enhance existing systems with a simpler and more elegant fetching mechanism between gateway and user context servers utilizing dependency injection design patterns.

2020 – 2022 **Graphics Hardware Engineer.** Intel.

- As the major contributor, developed a novel, patentable, fixed bitwidth constant modulus hardware architecture. Casted the fixed bitwidth constant modulus algorithm as a linear system with constrained variables to prove the feasibility of this hardware design. Implemented Register Transfer Level (RTL) generator in Python for lookup table based methods for experimental comparison. Drafted research paper manuscript for journal submission and patent application.
  This hardware component was adopted by multiple hardware designs in the Intel graphics division and also resulted in being a recipient of the Intel Distinguished Invention Award.
- Designed and implemented a parameterizable floating point adder in SystemVerilog as well as other components such as adders, multipliers, modulus hardware. This was to create inhouse RTL components competitive with Designware's suite of commercially distributed plug-and-play designs.
- Excavated, presented, and suggested improvements for RTL designs in graphics hardware pipelines owned by sister teams.
- Developed waterfall proof flows to carry out formal verification of RTLs with Hector formal verification engine with TCL scripts.
- Developed push button synthesis flows to generate area/delay pareto efficient comparisons for multiple RTLs using Python.
- Analyzed and presented various parts of floating point accumulator (systolic) pipeline to other teams within the graphics department.
- Explored procedural operation trees as policies for explainability in reinforcement learning.
- Presented various research papers as digestable powerpoints and created concise presentations for documentation purposes

Employment (continued)

2019 Summer — **Software Engineer Intern.** American Express.

- Made integrations generating Jira tickets from group messages for company Slack.

Education

2016 – 2020 — **B.Sc. Mathematics and Computer Science** University of California, San Diego.

Skills

Mathematics	Strong ability at formulating mathematical solutions for engineering problem solving with successful professional experience.
Software Languages	Strong Professional experience in software development using Typescript, C++, Java, Python; in RTL implementation using SystemVerilog; and in formal verification scripting using TCL.
Databases and Storage	Strong experience in relational databases and data mirroring technology application.
Serverside development	Strong working experience in serverside development using technologies including Remote Procedure Call (RPC) frameworks, caching systems, offline data pipelines, and event-driven architectures (Pub/Sub).

Awards & Additional Experiences

2021 — **Merit Award**, Intel Distinguished Invention Award.

2019 — **Reinforcement Learning Project**
Explored using imitation learning algorithm to train simulation sawyer robot to pick up cube from demonstrations. This was done while volunteering at the UCSD ARCLAB.

TA for Undergraduate Class in Mathematics for Algorithms and Systems
Teaching Assistant for undergraduate class for discrete mathematics. The focus of this class was proof methods, digital circuit logic, and basic graph theory.